Data Structures

ASSIGNMENT 4

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PART - A

Q1. Breadth First Search (BFS)

Starting Vertex: A

Steps

1. Start BFS from vertex A
2. Mark A as visited and insert into queue
3. Visit all unvisited adjacent vertices level by level

Traversal Order

A → B → C → D → E → F → G → H

Queue Movement

- Start: [A]
- After A: [B, C]
- After B: [C, D]
- After C: [D, E]
- After D: [E, F]
- After E: [F, G]
- After F: [G, H]
- End: [ ]

Levels / Hops from A

- A → Level 0 → 0 hops
- B, C → Level 1 → 1 hop
- D, E → Level 2 → 2 hops
- F, G → Level 3 → 3 hops
- H → Level 4 → 4 hops

Q2. Depth First Search (DFS)

Starting Vertex: A

Steps

1. Start DFS from A
2. Visit one vertex deeply before backtracking
3. Continue until all vertices are visited

DFS Traversal

A → B → C → D → E → F → H → G

Tree Edges

A-B, B-C, C-D, D-E, E-F, F-H, H-G

Back Edges

C-A, D-B, E-C

PART - B

Q3. Kruskal's Algorithm

Steps

1. Sort all edges in increasing order of weight
2. Select the smallest edge
3. If it forms a cycle → reject
4. Repeat until V - 1 edges are selected

Selected Edges

- B-C (1)
- B-D (2)
- E-F (2)
- G-H (2)
- A-C (3)
- D-E (3)
- F-H (3)

Rejected Edges

- A-B, C-D, E-G, D-F, C-E

(They form cycles)

Q4. Prim's Algorithm

Starting Vertex: A

Steps

1. Start from vertex A
2. Select minimum weight edge connected to visited set
3. Add new vertex to MST
4. Repeat until all vertices are included

Selected Edges

A-C, C-B, B-D, D-E, E-F, F-H, H-G

Total Cost:

16

Comparison

- Prim's MST cost = 16
- Kruskal's MST cost = 16

PART - C

Q5. Dijkstra's Algorithm

Source Vertex: A

Steps

1. Assign distance 0 to A and ∞ to others
2. Pick vertex with minimum distance
3. Relax all adjacent edges
4. Repeat until all vertices are visited

Final Distances from A

- A → 0
- C → 3
- B → 4
- D → 6
- E → 9
- F → 11

- G → 13
- H → 14

Shortest Path (A → H)

A → B → D → E → F → H

Total Travel Time:

14 minutes